

Programación de microcontroladores

GPIO: entradas y salidas

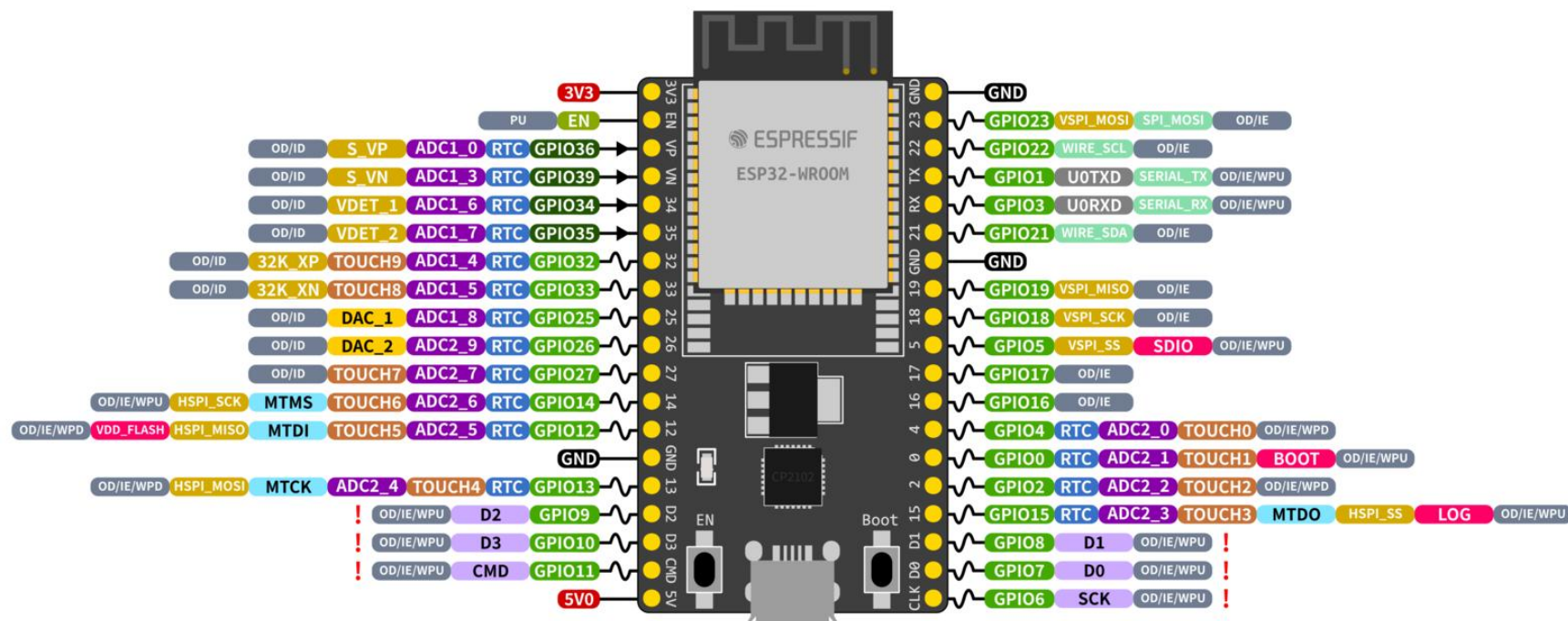
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Pines GPIO

General Purpose Input/Output.

Son los pines físicos del microcontrolador que se pueden configurar como:

- **Entrada:** leer señales.
- **Salida:** enviar señales.



ESP32 Specs

32-bit Xtensa® dual-core @240MHz
 Wi-Fi IEEE 802.11 b/g/n 2.4GHz
 Bluetooth 4.2 BR/EDR and BLE
 520 KB SRAM (16 KB for cache)
 448 KB ROM
 34 GPIOs, 4x SPI, 3x UART, 2x I2C,
 2x I2S, RMT, LED PWM, 1 host SD/eMMC/SDIO,
 1 slave SDIO/SPI, TWAI®, 12-bit ADC, Ethernet



RTC (Blue): RTC Power Domain (VDD3P3_RTC)
GND (Black): Ground
PWD (Red): Power Rails (3V3 and 5V)
! (Red): Pin Shared with the Flash Memory
 Can't be used as regular GPIO

GPIO STATE

- WPU:** Weak Pull-up (Internal)
- WPD:** Weak Pull-down (Internal)
- PU:** Pull-up (External)
- IE:** Input Enable (After Reset)
- ID:** Input Disabled (After Reset)
- OE:** Output Enable (After Reset)
- OD:** Output Disabled (After Reset)

Manejo de pines GPIO



```
1 #include "freertos/FreeRTOS.h"  
2 #include "freertos/task.h"  
3 #include "driver/gpio.h"
```

GPIO como salida

```
1 gpio_config_t io_conf = {  
2  
3     .pin_bit_mask = (1UL << PIN_NUMBER),  
4     .mode = GPIO_MODE_OUTPUT,  
5     .pull_up_en = GPIO_PULLUP_DISABLE,  
6     .pull_down_en = GPIO_PULLDOWN_DISABLE,  
7     .intr_type = GPIO_INTR_DISABLE  
8  
9 };  
10  
11 gpio_config(&io_conf);
```

GPIO como entrada

```
1 gpio_config_t io_conf = {  
2     .pin_bit_mask = (1ULL << BUTTON_PIN),  
3     .mode = GPIO_MODE_INPUT,  
4     .pull_up_en = GPIO_PULLUP_ENABLE,  
5     .pull_down_en = GPIO_PULLDOWN_DISABLE,  
6     .intr_type = GPIO_INTR_DISABLE  
7 };  
8  
9 gpio_config(&io_conf);
```

Configurar niveles de salida de un pin



```
1 gpio_set_level(LED_PIN, 1); // Salida en alto(3.3V)
2 gpio_set_level(LED_PIN, 0); // Salida en bajo (GND)
```

Retardos (bloqueantes)



```
1 vTaskDelay(pdMS_TO_TICKS(500)); // Valor en ms
```


Lectura del nivel de un GPIO

```
1 bool state = gpio_get_level(BUTTON_PIN);
```